WORKSHEET

Chapter 6: Remainder and factor theorems

1 Given that $m(z) = (5 + i)z^2 - (3 - 4i)z + 8 - 11i$, find the remainder when $m(z)$ is divided by each of the following.

a $z + 1$

b $z - i$

c $z + 3i$

d $z - 2 + 3i$

e $z + 4 - 3i$

f $z + 5 - 4i$

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2 Given that $p(z) = (1 + 3i)z^3 + (2 - 5i)z^2 - (3 + i)z - 2 + 6i$, find the remainder when $p(z)$ is divided by each of the following.

a $z + 2$

b $z - 3i$

c $z + 2i$

d $z + 4 - 3i$

e $z - 2 + i$

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f $z + 1 - 3i$

3 Given that $q(z) = (1 - 2i)z^4 + 3z^2 - (5 + 4i)z + 7 - 8i$, find the remainder when $q(z)$ is divided by each of the following.

a $z - 3i$

b $z + 1 - 6i$

c $z - 3 - i$

d $z + 2 + 5i$

e $z - 4 + 5i$

f $z - 2 - 3i$

4 Given that $r(z) = 2z^5 - 3iz^4 + (5 - 2i)z^3 - 4iz^2 + 7z - 11 + 9i$. Find the remainder when $r(z)$ is divided by each of the following.

a $z + 3i - 4$

b $z - i + 3$

c $z - 2 + 3i$

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d $z + 5 + 2i$

e $z + 2 - i$

f $z - 3 + 2i$

5 Show that $(z + 3i)$ is a factor of $p(z) = (2 + i)z^3 + (-8 + 12i)z^2 - (11 + 18i)z + 9 + 21i$.

6 Show that $(z - 2 + 3i)$ is a factor of $p(z) = (3 + 2i)z^3 - (7 + 2i)z^2 + (7 + 32i)z - 1 - 18i$.

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7 Show that $(z - 2)$ is a factor of $p(z) = 4iz^4 + (8 - 11i)z^3 + (-19 + 8i)z^2 + (13 - 9i)z - 14 + 10i$.

8 Show that $(z + 4 - 3i)$ is a factor of $p(z) = (2 - 3i)z^5 - (2 + 20i)z^4 - (17 + 10i)z^3 - (19 + 22i)z^2 + (9 + 37i)z + 21 - 72i$.

9 Show that $(z + 2)$, $(z - 3i)$ and $(z + 2 - 3i)$ are all factors of $p(z) = z^3 + (5 + 6i)z - 12 - 18i$.

10 Show that $(z - 2)$, $(z + 3i)$ and $(z - 3 + 4i)$ are all factors of $p(z) = z^3 + (-5 + 7i)z^2 - (6 + 23i)z + 24 + 18i$.

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Answers

1 **a** $35 - 5i$

b $28 - 4i$

c $44 + 8i$

d $38 + 16i$

e $32 - 26i$

f $31 - 33i$

2 **a** $4 - 36i$

b $64 + 15i$

c $-36 + 40i$

d $-400 + 167i$

e $12 - 21i$

f $38 + 86i$

3 **a** $73 - 185i$

b $2692 - 1392i$

c $291 - 536i$

d $-1705 - 837i$

e $-139 + 3639i$

f $-365 + 123i$

4 **a** $-7773 - 1518i$

b $-358 + 698i$

c $311 + 1606i$

d $4895 - 9356i$

e $-25 + 166i$

f $-1729 - 124i$

5–10 Show $p(a) = 0$ [where $(z - a)$ is the factor of $p(z)$] in each case.

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